

CANONICAL LESSON

See the Service Gaps: Informal-Settlement Mapping

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C53
Notebook	V7_D_N03
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Map service gaps in informal settlements using accessible, privacy-preserving, participatory evidence that supports equitable investment without exposing or stigmatizing residents.

Learning outcomes

Why this matters

High-resolution maps can expose vulnerable households, reinforce stigma, or become instruments of eviction rather than service improvement.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

Residents and service agencies validate aggregated water, sanitation, access, drainage, and waste gaps before prioritization.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Replace straight-line distance with a network measure.
- 2.

- Add uncertainty ranges to coverage estimates.
- 3.
- Design a community-validation protocol.
- 4.
- Explain why “informal” is not an analytical risk label.

Answer key

- 1.
- Use an authoritative routable network, travel modes, barriers, time, accessibility, and CRS validation.
- 2.
- Propagate survey/sensor/classification uncertainty and show interval-sensitive ranks.
- 3.
- Define representation, consent, accessible communication, correction, grievance, feedback, and no-retaliation safeguards.
- 4.
- Informality is a legal/social planning context; converting it into a risk score can stigmatize residents and enable punitive misuse.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.